Uncertainty-Aware Hierarchical Multi-Agent Reinforcement Learning with Graph-Based CTDE Critics and Topology-Conditioned Option Termination for Scalable Search-and-Rescue Swarms

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1 Abstract

Search-and-rescue (SAR) operations require decentralized multi-agent systems capable of long-horizon coordination, scalable interaction modeling, and adaptive task allocation under uncertainty. While hierarchical multi-agent reinforcement learning (MARL) improves temporal abstraction, existing approaches typically rely on fixed reward shaping, unstructured centralized critics, and simplistic option termination mechanisms, limiting scalability and robustness.

We propose an uncertainty-aware hierarchical MARL framework under centralized training with decentralized execution (CTDE), integrating (1) entropy-driven potential-based high-level rewards for information-efficient exploration, (2) graph-based centralized critics respecting interaction topology, and (3) graph-conditioned option termination that adapts skill switching to local swarm structure.

Our formulation models SAR as an information-theoretic objective minimizing belief entropy over victim locations. We demonstrate that graph-based critics improve scalability, potential-based rewards prevent reward hacking, and topology-conditioned termination improves throughput in constrained navigation. Experimental evaluation across increasing swarm sizes and bottleneck scenarios validates improved coordination efficiency, convergence stability, and scalability over flat and non-graph hierarchical baselines.

2 Introduction

Multi-agent search-and-rescue (SAR) tasks involve distributed robots operating under partial observability to locate victims, explore unknown terrain, and co-

ordinate under communication and safety constraints. Such environments are characterized by:

- Long-horizon temporal dependencies,
- Dynamic interaction graphs,
- Uncertain and evolving belief states,
- Non-stationarity induced by multiple learning agents.

Flat multi-agent reinforcement learning (MARL) approaches struggle with scalability and credit assignment as agent count increases. Hierarchical MARL introduces temporal abstraction via options, improving long-horizon reasoning. However, three limitations remain:

1. High-level reward functions often rely on handcrafted shaping prone to reward hacking.
2. Centralized critics typically ignore interaction graph structure, limiting scalability.
3. Option termination mechanisms do not account for local topology, leading to oscillatory switching or congestion in constrained environments.

To address these limitations, we propose a unified framework integrating:

- Information-theoretic high-level objectives via entropy reduction,
- Graph-based centralized critics respecting permutation invariance,
- Graph-conditioned option termination driven by local interaction structure.

The key contributions are:

- A belief-entropy potential-based hierarchical reward for SAR.
- A graph-based centralized critic under CTDE for scalable coordination.
- A topology-conditioned option termination mechanism improving stability and throughput.

3 Related Work

3.1 Hierarchical Multi-Agent Reinforcement Learning

Hierarchical MARL frameworks decompose control into high-level task selection and low-level execution. ALMA demonstrates task allocation and execution factorization for composite multi-agent tasks. However, allocation is not uncertainty-driven and critics do not leverage interaction topology.

TAG introduces multi-level hierarchical abstractions but does not incorporate information-theoretic objectives or graph-conditioned option termination.

Our approach differs by integrating entropy-based allocation and graph-aware critics within hierarchical MARL.

3.2 Graph Neural Networks in MARL

Graph neural networks (GNNs) have been used for decentralized policies due to permutation equivariance and scalability. Most prior work applies GNNs to the actor only.

We extend this paradigm by incorporating a graph-based centralized critic, improving inductive bias and scalability in CTDE settings.

3.3 Option-Critic and Termination Learning

Option-Critic learns termination functions end-to-end. However, termination decisions are typically conditioned only on local features. We introduce topology-conditioned termination via GNN embeddings, enabling context-aware switching behavior in congested or structured environments.

3.4 Information-Theoretic Exploration

Entropy-based exploration objectives are common in active information gathering. We formalize SAR as minimizing belief entropy and incorporate this directly into hierarchical reward shaping via potential differences.

4 Methods

4.1 Problem Formulation

We consider a Dec-POMDP:

$$\mathcal{M} = \langle \mathcal{S}, \{\mathcal{O}_i\}, \{\mathcal{A}_i\}, P, r, \gamma \rangle$$

Agents maintain a belief map:

$$b_t(x) = P(\text{victim at } x \mid \mathcal{H}_t)$$

Belief entropy:

$$H(b_t) = -\sum_x b_t(x) \log b_t(x)$$

Mission potential:

$$\Phi(s_t) = -H(b_t)$$

High-level reward:

$$R_k^H = \Phi(s_{t_{k+1}}) - \Phi(s_{t_k}) - \lambda_t(t_{k+1} - t_k) - \lambda_{sw}\, \mathbf{1}[z_k \neq z_{k-1}]$$

4.2 Hierarchical Policy Structure

At decision times t_k:

$$z_k^i \sim \pi_H^i(z \mid h_{H,k}^i)$$

Low-level execution:

$$a_t^i \sim \pi_L^i(a \mid h_{L,t}^i, z_k^i)$$

Options follow the SMDP formulation:

$$Q(s, z) = \mathbb{E}\big[R(s, z, k) + \gamma^k V(s')\big]$$

4.3 Graph Construction

Interaction graph:

$$G_t = (V, E_t)$$

Edges are defined by a proximity threshold.
Node features:

$$x_i = [o_i, z_i]$$

Edge features include relative displacement.

4.4 GNN Actor

Message passing:

$$m_{ij} = \phi(h_i, h_j, e_{ij})$$

$$m_i = \sum_{j \in \mathcal{N}(i)} m_{ij}$$

$$h_i' = \psi(h_i, m_i)$$

Policy:

$$\pi_L^i(a_i \mid h_i')$$

4.5 Graph-Based Centralized Critic

Global message passing:

$$h_i^{(K)} = \mathrm{GNN}_\psi(x_i, \{x_j\})$$

Pooling:

$$h_{\text{global}} = \mathrm{POOL}(\{h_i^{(K)}\})$$

Value estimate:

$$V_\psi(s_t) = f(h_{\text{global}})$$

Advantage:

$$A_t = \hat{R}_t - V_\psi(s_t)$$

4.6 Graph-Conditioned Option Termination

Termination probability:

$$\beta_t^i = \sigma\Big(\mathrm{MLP}\Big(\mathrm{GNN}_\beta(o_t^i, \{o_t^j\})\Big)\Big)$$

Constraints:

$$k_{\min} \le t - t_k \le k_{\max}$$

Termination sampling:

$$\text{terminate if } u < \beta_t^i$$

This enables topology-aware switching.

4.7 Proper Mission-Level Potential Formulation

Instead of defining

$$\Phi(s) = -H(b),$$

we introduce a phase-aware potential function:

$$\Phi(s) = w_E \Phi_E(s) + w_N \Phi_N(s) + w_F \Phi_F(s).$$

Exploration Term

$$\Phi_E(s) = -H(b).$$

Navigation Term Distance-to-goal formulation:

$$\Phi_N(s) = -\sum_i d_i^{\text{goal}}.$$

Alternatively, bottleneck throughput proxy:

$$\Phi_N(s) = \#\text{agents past choke line}.$$

Formation Term

$$\Phi_F(s) = -\sum_i \left(\|x_i - c\| - R\right)^2 - \sum_i \left(\Delta\theta_i - \frac{2\pi}{N}\right)^2.$$

High-Level Reward The high-level reward becomes:

$$R_k^H = \Phi(s_{t_{k+1}}) - \Phi(s_{t_k}) - \lambda_t \Delta t - \lambda_{sw}\, \mathbf{1}[z_k \neq z_{k-1}].$$

5 Problem Formulation

We consider a cooperative multi-agent partially observable Markov decision process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{S}, \{\mathcal{O}_i\}_{i=1}^N, \{\mathcal{A}_i\}_{i=1}^N, P, r, \gamma, N \rangle$$

- N agents
- Global state $s \in \mathcal{S}$
- Local observations $o_i \in \mathcal{O}_i$
- Actions $a_i \in \mathcal{A}_i$
- Transition $P(s' \mid s, a)$
- Discount γ

Agents operate in a search-and-rescue (SAR) environment with unknown victim locations.

Belief Map and Uncertainty Objective

$$b_t(x) = P(\text{victim at } x \mid \mathcal{H}_t)$$

$$H(b_t) = -\sum_x b_t(x) \log b_t(x)$$

Define mission potential:

$$\Phi(s_t) = -H(b_t)$$

Objective:

$$\max_\pi \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \big(\Phi(s_{t+1}) - \Phi(s_t)\big)\right]$$

This is equivalent to maximizing cumulative entropy reduction.

6 Hierarchical Architecture

We model the system as a Semi-Markov Decision Process (SMDP) with options.

Each option:

$$o = (\mathcal{I}_o, \pi_o, \beta_o)$$

6.1 High-Level Policy

At decision times t_k:

$$z_k^i \sim \pi_H^i(z \mid h_{H,k}^i)$$

High-level reward:

$$R_k^H = \Phi(s_{t_{k+1}}) - \Phi(s_{t_k}) - \lambda_t(t_{k+1} - t_k) - \lambda_{sw}\mathbf{1}[z_k \neq z_{k-1}]$$

6.2 Low-Level Policy

At each timestep:

$$a_t^i \sim \pi_L^i(a \mid h_{L,t}^i, z_k^i)$$

6.3 Option-Conditioned Potential

If a global option z is selected, we define the potential as:

$$\Phi(s, z) = \begin{cases} -H(b), & z = \text{Explore}, \\ -\sum_i d_i^{\text{goal}}, & z = \text{Navigate}, \\ -\text{FormationError}, & z = \text{Form}. \end{cases}$$

Manager Behavior The high-level manager learns:

- When entropy is high → **Explore**
- After victim detected → **Navigate**
- Near victim → **Form**

7 Graph-Based Actor and Critic (CTDE)

7.1 Interaction Graph

$$G_t = (V, E_t)$$

Nodes correspond to agents. Edges exist if $\|p_i - p_j\| \leq r$.
Node features:

$$x_i = [o_i, z_i]$$

Edge features:

$$e_{ij} = [p_j - p_i, \|p_j - p_i\|]$$

7.2 GNN Actor (Decentralized)

Message passing:

$$m_{ij} = \phi_\theta(h_i, h_j, e_{ij})$$

$$m_i = \sum_{j \in \mathcal{N}(i)} m_{ij}$$

$$h'_i = \psi_\theta(h_i, m_i)$$

Policy:

$$\pi_L^i(a_i \mid h'_i)$$

Permutation equivariant by construction.

7.3 Graph-Based Centralized Critic

$$h_i^{(K)} = \mathrm{GNN}_\psi(x_i, \{x_j\})$$

Global pooling:

$$h_{\text{global}} = \mathrm{POOL}(\{h_i^{(K)}\})$$

Value estimate:

$$V_\psi(s_t) = f_\psi(h_{\text{global}})$$

Advantage:

$$A_t = \hat{R}_t - V_\psi(s_t)$$

This enforces permutation invariance and scales with $\mathcal{O}(|E|)$.

8 Graph-Conditioned Option Termination

Traditional termination:

$$\beta_t^i = \sigma(\mathrm{MLP}(o_i))$$

Proposed termination:

$$\beta_t^i = \sigma\Big(\mathrm{MLP}\left(\mathrm{GNN}_\beta(o_t^i, \{o_t^j\}_{j \in \mathcal{N}(i)})\right)\Big)$$

Constraints:

$$k_{\min} \leq t - t_k \leq k_{\max}$$

9 Full Learning Objective

Low-level MAPPO objective:

$$L_L = \mathbb{E}\left[\min\left(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t\right)\right]$$

Where:

$$r_t(\theta) = \frac{\pi_{L,\theta}(a_t \mid h_t, z)}{\pi_{L,\theta_{\text{old}}}(a_t \mid h_t, z)}$$

High-level PPO objective defined analogously using R_k^H.
Termination objective:

$$L_\beta = \mathbb{E}\left[\beta_t \cdot A_t^H\right]$$

10 Methodological Adaptations

Paper	Adopted Concept	Extension
ALMA	Hierarchical allocation	Entropy-based potential
TAG	Multi-level graph abstraction	Graph critic + GNN actors
HMARL-CBF	Hierarchical safety	Optional integration
Learned Shields	Decentralized filtering	Future extension
UBSRL	Event-triggered control	Combined with termination

11 Experimental Design

11.1 Scenarios

- Large-area exploration
- Multi-victim clustered environment
- Narrow passage bottleneck
- Scaled agents: $5 \to 20 \to 40$

11.2 Baselines

- Flat MAPPO (MLP critic)
- Hierarchical without graph critic
- Hierarchical without potential reward
- Hierarchical without graph termination

11.3 Metrics

- Entropy reduction rate
- Time-to-detection
- Redundant coverage overlap
- Bottleneck throughput
- Switching frequency
- Scalability vs. N

12 Hypotheses

- H1: Graph critic improves scalability and convergence.
- H2: Potential-based reward prevents reward hacking and improves entropy reduction.
- H3: Graph-conditioned termination reduces deadlock and improves passage throughput.
- H4: No big distributional shift is observed across training and evaluation(no maze-like obstacles during testing phase; when training had open spaces)

13 Algorithm Summary

1. Initialize belief map.
2. Pretrain low-level skills.
3. Train hierarchical system with:
    - Graph actor
    - Graph critic
    - Potential reward
    - Graph termination
4. Evaluate against baselines.

Algorithm 1 Uncertainty-Aware Hierarchical MARL with Graph Critic and Topology-Conditioned Termination

Initialize low-level policy π_L, high-level policy π_H
Initialize graph critic V_ψ
Initialize termination network β
Initialize belief map b
for episode $= 1$ to M **do**
  Reset environment
  Compute initial belief map b_0
  $t \leftarrow 0$
  while not done **do**
    if option terminated **or** $t = 0$ **then**
      for each agent i **do**
        Compute high-level embedding h_H^i
        Sample option $z_i \sim \pi_H(z \mid h_H^i)$
      end for
      $t_k \leftarrow t$
    end if
    for step $= 1$ to K **do**
      Build interaction graph G_t
      Compute GNN embeddings h_L^i
      for each agent i **do**
        Sample action $a_i \sim \pi_L(a \mid h_L^i, z_i)$
      end for
      Execute joint action
      Observe next State s_{t+1}
      Update belief map b_{t+1}
      Store transition
      Compute termination probability:

$$\beta_i = \sigma(\mathrm{MLP}(\mathrm{GNN}_\beta(o_i, \text{neighbors})))$$

      if termination condition met **then**
        break
      end if
      $t \leftarrow t + 1$
    end for
    Compute high-level reward:

$$R_H = \Phi(s_{t+1}) - \Phi(s_t) - \lambda_{\text{time}} \Delta t - \lambda_{\text{switch}} \, \mathbf{1}[\text{switch}]$$

  end while
  Compute advantages using graph critic V_ψ
  Update π_L via MAPPO
  Update π_H via PPO
  Update termination network β
  Update critic V_ψ
end for